

MLB Scion — Task and Model Summary

Two LightGBM ensemble classes for home-team win classification

1. The task

The objective is to develop a machine learning-based system to predict the outcome of Major League Baseball (MLB) games that fall within the bookie price range of +/-150. More specifically, the models classify **whether or not the home team will win** each game. This is a binary classification problem (home win = 1, home loss = 0); the model's home-win probability drives downstream bet selection, edge measurement against the bookmaker's price, and stake sizing.

2. Two classes of model ensemble

Two distinct classes of gradient-boosted (LightGBM) ensemble predict the same target. They share identical training and evaluation data and differ only in their **feature inputs**, which is what makes their predictions usefully decorrelated.

- **V119 — line + stats (15-model ensemble).** Team and pitcher performance statistics together with the bookmaker's market information for the game: opening line, bet line, the difference between them (line movement), and historical bookmaker probabilities.
- **V123 — stats-led, game line removed (41-model ensemble).** The same style of statistics but with the quoted opening and bet line for the game removed (and the bet-vs-open difference). The vacated slots are filled with further high-ranking statistics — notably bullpen workload/effectiveness (BallpenOuts, BallpenERA) and additional men-on-base and pitcher-strikeout measures.

3. Training and evaluation data (identical for every model)

- **In-sample (training):** MLB seasons 2004 up to 5 June 2025 (only games with bookie line range +/-150.)
- **Out-of-sample (evaluation):** 6 June 2025 to 5 June 2026 (1,770 unseen games where bookie line range +/-150).
- **Individual LGBM models are only allowed into the ensemble if they pass a threshold level performance on the Out-of-sample** (eg >52% for either Precision or Recall and that F1 must be >51%)
- **As a result, after 2 weeks spent generating LGBM trees, we have a V119 (Line and Stats) LGBM ensemble of 15 trees (selected from 2,500), and a V123 (StatsOnly) LGBM ensemble of 41 trees (selected from 1,500)** — clearly it is easier for the LGBM to find more useful trees for V123 (StatsOnly) than V119. Note that a results spreadsheet has been developed that includes all LGBM voters, V119 ensemble performance, V123 ensemble performance and V119 and V123 agreement/consensus performance. The BEST probability points edge (difference between the model's home probability and the bookie's devigged home probability) is determined using a control box in the spreadsheet, where we enter different thresholds and review impact on ROI etc) and a play is ruled out if there is no edge

found (we never play against the original classification). The threshold of V119 ensemble is 0.006 and the threshold for V123 is 0.008. Also, we establish the minimum acceptable agreement threshold for V119 is 0.53 and the minimum voter agreement for V123 is 0.51. Now this might be viewed as a risk of fitting to the outsample but we consider this minimal. The acid test will be when we put the model into production for games beyond the outsample ie from 6th Jun 2026 onwards. Note: Six of the Seven macStudios are now running on V119 data to search for stronger models.

- **The strategy applied based on the companion document**
MLB_Scion_V26_06a_Implementation_Plan_v4_18Jun2026.docx is as follows:

High Stake Plays:

V119 and V123 Consensus play on Dogs

Mid Stake Plays:

V119 play on Faves

Low Stake Plays:

V119 and V123 Consensus play on Faves

4. Ensemble-level feature profiles (SHAP)

Aggregated across all members of each ensemble (14 of 15 V119 models; all 41 V123 models). **Share %** = mean contribution to total importance; **Mean rank** = average importance rank (1 = top); **Rank SD** = standard deviation of that rank across models (lower = more consistent). **Stability tier** classifies each feature (see §5). All features shown have full coverage (used by every model).

V119 — ensemble top 10

#	Feature	Share %	Mean rank	Rank SD	Stability tier
1	G_Bookie_H_Probability	24.8%	1.0	0.00	Core
2	G_VHRatio_DaysRest	4.9%	8.5	10.34	Variable
3	G_VHRatio_ClosingProbabilityLine_HV	4.1%	7.9	7.08	Variable
4	V_StartingPitcher_StrikeoutAccuracy_YTD	3.8%	5.1	3.65	Steady
5	G_VHRatio_MenOnBase_Strength_YTD	3.7%	9.4	8.46	Variable
6	G_VHRatio_Runs_5InningsAllowed_YTD	2.9%	9.6	8.77	Variable
7	G_VHRatio_MenOnBase_Allowed_20G	2.8%	9.6	5.42	Steady
8	V_MenOnBase_Strength_YTD	2.8%	11.2	8.13	Variable
9	G_VHRatio_StartingPitcher_Strikes_Ratio_5G	2.6%	10.9	7.14	Variable
10	G_VHRatio_K_Per_9_YTD	2.3%	15.1	6.74	Variable

The bookmaker's line is V119's anchor: **24.8% of importance, #1 in all 14 models with zero rank variance (the sole Core feature)**. By contrast *G_VHRatio_DaysRest* is 2nd by share but **Variable** (SD 10.34, ranging 2nd–35th) — an occasional strong signal, not a steady driver; share alone would mislead.

V123 — ensemble top 10

#	Feature	Share %	Mean rank	Rank SD	Stability tier
1	G_VHRatio_MenOnBase_Strength_YTD	14.6%	1.4	0.83	Core
2	G_VHRatio_ClosingProbabilityLine_HV	11.2%	1.8	0.59	Core
3	V_StartingPitcher_StrikeoutAccuracy_YTD	6.9%	4.1	2.03	Core
4	G_VHRatio_StartingPitcher_StrikeoutsImpact_YTD	6.1%	4.8	2.55	Steady
5	H_BallpenOuts_YTD	6.0%	4.7	1.67	Core
6	V_ClosingProbabilityLine_YTD_HV	3.4%	10.0	6.56	Variable
7	G_VHRatio_StartingPitcher_StrikesImpact_YTD	3.1%	12.3	7.82	Variable
8	G_VHRatio_OPS_YTD	2.9%	11.7	5.76	Steady
9	G_VHRatio_MenOnBase_Allowed_20G	2.9%	10.9	5.66	Steady
10	G_VHRatio_StartingPitcher_WalkHitsAllowedPerInning_5G	2.8%	11.5	4.45	Steady

With the game line removed, V123 rests on **four Core fundamentals** used near the top of every model — men-on-base strength, a closing-line ratio, starting-pitcher strikeout accuracy, and bullpen outs. *G_Bookie_H_Probability* is absent from every V123 model.

5. Driver stability — tiers and what they tell us

Each feature is classified into a stability tier from its rank consistency across models (these tiers are included in the accompanying *V119_ensemble_SHAP.csv* and *V123_ensemble_SHAP.csv* files):

- **Core** — rank SD ≤ 2.5 and used by every model: lands near the top consistently (a dependable driver).
- **Steady** — rank SD ≤ 6 and $\geq 80\%$ coverage: reasonably consistent.
- **Variable** — otherwise: importance swings model-to-model (an occasional signal).

The tier counts capture the two ensembles' characters in one line:

- **V119: 1 Core, 6 Steady, 31 Variable** — a single dominant Core driver (the market line) with a largely Variable supporting cast.
- **V123: 4 Core, 20 Steady, 11 Variable** — no single dominator, but a broad, stable base of fundamentals.

Why the difference. In V119 the market line absorbs ~25% of importance, so the statistics compete over a small residual and are partly **substitutable** — different models lean on different stats, hence mostly Variable ranks (the steadiest stat is starting-pitcher strikeout accuracy, SD 3.65). In V123, with no line to lean on, the models **converge on the same genuinely predictive fundamentals**, which become Core precisely because they are load-bearing. V123's larger ensemble (41 vs 14) aids stability, but the very low per-feature SDs reflect real agreement, not just sample size. This stable fundamental core is the independent read that, when it agrees with V119's line-anchored view, produces the high-conviction consensus.

One Core feature in V123 is itself market-derived — the closing-line ratio (rank SD 0.59, always top-3) — which leads into the nuance below.

6. A nuance: V123 is not fully market-blind

V123's second-largest and most-stable driver is a **closing-line ratio** (*G_VHRatio_ClosingProbabilityLine_HV*, 11.2%, Core), with another closing-line feature lower down. So V123 excludes the game's *opening and bet* line but retains **closing-line-derived ratio features**, which are market-derived. The two ensembles are therefore best described as weighting market and fundamental information **very differently** (V119 led by the line; V123 led by fundamentals with a residual closing-line signal) rather than as "market-aware vs market-blind". Worth confirming whether *ClosingProbabilityLine* derives from the target game's closing price or from historical team context — it refines how independent V123 truly is, though not the empirical consensus result.

7. How the ensembles are used together

Evaluated three ways out-of-sample: V119 alone, V123 alone, and in **agreement (consensus)** — betting only when both independently back the same side. Because the two weight market and fundamental information so differently, their agreement combines two largely independent reads, and the consensus subset is materially stronger than either alone. Full performance, the information-content decomposition, and the staking policy are in the companion document *MLB_Scion_V26_06a_Implementation_Plan_v4_18Jun2026.docx*.

This can be summarised as:

High Stake Plays:

V119 and V123 Consensus play on Dogs

Mid Stake Plays:

V119 play on Faves

Low Stake Plays:

V119 and V123 Consensus play on Faves

8. The June Problem

For some reason, the month of June in the MLB season seems to be a very low yield and costly month.

This can be seen in the Jun 2025 outsample performance for both models independently and when they agree. Below for each combination (V119 only, V123 only, and consensus plays between V119 and V123), there is the overall performance for the outsample (6th Jun 2025 to 5th Jun 2026), showing a breakdown of Faves (F) and Dogs (D). After the overall breakdown, we then provide such a performance breakdown for 2025 over all, followed by Jun 2025 only; we then show performance for 2026 (til 5th Jun). Please see tables a) to p) below.

a)V119 ensemble – full outsample

	Full Outsample
#plays	357
%totalGames	20.2%
#wins	204
%winRate	57.1%
uInvestd	357.00
uRetnd	393.95
%ROI	10.3%
Units	36.95
SD(ROI)	96.2%
SR	
(outsample)	2.03
SR	
(playbyplay)	0.11

Fave and Dogs breakdown:

F	163.00	9.2% plays	D	194.00	11.0% plays
	178.29			215.66	
	ROI 9.4%	Units 15.29		ROI 11.2%	Units 21.66

b)V119 ensemble – Season 2025

	2025
#plays	202
%SeasonGames	19.2%
%OS_Games	11.4%
#wins	117
%winRate	57.9%
uInvestd	202.00
uRetnd	228.96
%ROI	13.3%
Units	26.96
	Full
	Outsample

Fave and Dogs breakdown:

F	77.00	38.1%	D	125.00	61.9%
	89.77			139.20	
	ROI 16.6%	Units 12.77		ROI 11.4%	Units 14.20

c)V119 ensemble – Jun 2025

	Jun
#plays	44
%totalGames	2.5%
#wins	25
%winRate	56.8%
uInvestd	44.00
uRetnd	47.28
%ROI	7.5%
Units	3.28

Fave and Dogs breakdown:

F			D		
	23.00	52.3%		21.00	47.7%
	29.17			18.11	
	26.8%	6.17		-13.7%	-2.89

d)V119 ensemble – Season 2026

	2026
#plays	155
%SeasonGames	21.5%
%OS_Games	8.8%
#wins	87
%winRate	56.1%
uInvestd	155.00
uRetnd	164.98
%ROI	6.4%
Units	9.98

Fave and Dogs breakdown:

F			D		
	86.00	55.5%		69.00	44.5%
	88.52			76.46	
	2.9%	2.52		10.8%	7.46

e)V123 ensemble – full outsample

	Full Outsample
#plays	526
%totalGames	29.7%
#wins	285
%winRate	54.2%
uInvestd	526.00
uRetnd	575.95
%ROI	9.5%
Units	49.95
SD(ROI)	101.2%
SR	
(outsample)	2.15
SR	
(playbyplay)	0.09
SR	
(outsample)	2.03

Fave and Dogs breakdown:

F	125.00	7.1%	D	401.00	22.7%
	137.18			438.77	
	9.7%	12.18		9.4%	37.77

f)V123 ensemble – Season 2025

	2025
#plays	299
%SeasonGames	28.5%
%OS_Games	16.9%
#wins	166
%winRate	55.5%
uInvestd	299.00

uRetnd	338.29
%ROI	13.1%
Units	39.29

Fave and Dogs breakdown:

F			D		
	60.00	20.1%		239.00	79.9%
	71.55			266.74	
	19.2%	11.55		11.6%	27.74

g)V123 ensemble – Jun 2025

	Jun
#plays	73
%totalGames	4.1%
#wins	34
%winRate	46.6%
uInvestd	73.00
uRetnd	68.83
%ROI	-5.7%
Units	-4.17

Fave and Dogs breakdown:

F			D		
	14.00	19.2%		59.00	80.8%
	18.77			50.06	
	34.1%	4.77		-15.2%	-8.94

h)V123 ensemble – Season 2026

	2026
#plays	227
%SeasonGames	31.5%
%OS_Games	12.8%
#wins	119
%winRate	52.4%
uInvestd	227.00
uRetnd	237.66
%ROI	4.7%
Units	10.66

Fave and Dogs breakdown:

	F		D	
	65.00	28.6%	162.00	71.4%
	65.63		172.03	
	1.0%	0.63	6.2%	10.03

i)V123 ensemble – outsample subset where bookie line -115 to 150

	Full Outsample
#plays	344
%totalGames	19.4%
#wins	205
%winRate	59.6%
uInvestd	344.00
uRetnd	414.16
%ROI	20.4%
Units	70.16
SD(ROI)	99.6%
SR (outsample)	3.80
SR (playbyplay)	0.20

Fave and Dogs breakdown:

	F		D	
	68.00	3.8%	276.00	15.6%
	86.13		328.03	
	26.7%	18.13	18.9%	52.03

j)V123 ensemble – season 2025 outsample subset where bookie line -115 to 150

	2025
#plays	201
%SeasonGames	19.1%
%OS_Games	11.4%
#wins	120
%winRate	59.7%
uInvestd	201.00
uRetnd	244.25
%ROI	21.5%
Units	43.25

Fave and Dogs breakdown:

F	35.00	17.4%	D	166.00	82.6%
	45.98			198.27	
	31.4%	10.98		19.4%	32.27

k)V123 ensemble – Jun 2025 outsample subset where bookie line -115 to 150

	Jun
#plays	54
%totalGames	3.1%
#wins	26
%winRate	48.1%
uInvestd	54.00
uRetnd	53.02
%ROI	-1.8%
Units	-0.98

Fave and Dogs breakdown:

F	10.00	18.5%	D	44.00	81.5%
	13.40			39.62	
	34.0%	3.40		-9.9%	-4.38

l)V123 ensemble – season 2026 (to 5th Jun) outsample subset where bookie line - 115 to 150

	2026
#plays	143
%SeasonGames	19.9%
%OS_Games	8.1%
#wins	85
%winRate	59.4%
uInvestd	143.00
uRetnd	169.91
%ROI	18.8%
Units	26.91

Fave and Dogs breakdown:

F	33.00	23.1%	D	110.00	76.9%
	40.14			129.76	
	21.6%	7.14		18.0%	19.76

m)V119 and V123 Ensemble Agreement– full outsample

	Full Outsample
#plays	320
%totalGames	18.1%
#wins	188
%winRate	58.8%
uInvestd	320.00
uRetnd	362.60
%ROI	13.3%
Units	42.60
SD(ROI)	95.7%
SR (outsample)	2.49
SR (playbyplay)	0.14

Fave and Dogs breakdown:

F	150.00	8.5%	D	170.00	9.6%
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166.85
11.2%
16.85

195.76
15.2%
25.76

n)V119 and V123 Ensemble Agreement– Season 2025

	2025
#plays	181
%SeasonGames	17.2%
%OS_Games	10.2%
#wins	108
%winRate	59.7%
uInvestd	181.00
uRetnd	211.21
%ROI	16.7%
Units	30.21

Fave and Dogs breakdown:

F	71.00	39.2%	D	110.00	60.8%
	84.01			127.20	
	18.3%	13.01		15.6%	17.20

o)V119 and V123 Ensemble Agreement– Jun 2025

	Jun
#plays	41
%totalGames	2.3%
#wins	24
%winRate	58.5%
uInvestd	41.00
uRetnd	45.31
%ROI	10.5%
Units	4.31

Fave and Dogs breakdown:

F	23.00	56.1%	D	18.00	43.9%
	29.17			16.14	
	26.8%	6.17		-10.3%	-1.86

p) V119 and V123 Ensemble Agreement– Season 2026

	2026
#plays	139
%SeasonGames	19.3%
%OS_Games	7.9%
#wins	80
%winRate	57.6%
uInvestd	139.00
uRetnd	151.40
%ROI	8.9%
Units	12.40

Fave and Dogs breakdown:

F			D		
	79.00	56.8%		60.00	43.2%
	82.83			68.56	
	4.9%	3.83		14.3%	8.56

8.1. Performance Beyond the Outsample: 6th June to 21st Jun 2026

The following shows performance on games (within bookie line +/- 150) for 6th Jun til 21st Jun 2026, highlighting significant performance issues. It is noticeable that the StatsOnly model is the only profitable model and therefore for 23rd Jun to 30th Jun we are considering switching to V123 StatsOnly model for our plays until July. NB these results are based on the closing line (some minor improvement is noticed on opening line and we usually bet on price mid way between).

a)V119 ensemble – 6th June to 21st Jun 2026

	#Games	208			
	#Plays	34			
	%Games	16.3%			
	#Win	15			
	%WinRate	44.1%			
	invest	3606			
	return	3100			
	profit	-506			
	%ROI	-14.0%			
HF	plays	14	VD	plays	16
	wins	7		wins	7
	%WR	50.0%		%WR	43.8%
	invest	1593		invest	1610

	return	1467		return	1431
	profit	-126		profit	-179
	%ROI	-7.9%		%ROI	-11.1%
VF	plays	0	HD	plays	4
	wins	0		wins	1
	%WR	0.0%		%WR	25.0%
	invest	0		invest	403
	return	0		return	202
	profit	0		profit	-201
	%ROI	0.0%		%ROI	-49.9%

b)V123 ensemble – 6th June to 21st Jun 2026

	#Games	208			
	#Plays	47			
	%Games	22.6%			
	#Win	24			
	%WinRate	51.1%			
	invest	4809			
	return	5045			
	profit	236			
	%ROI	4.9%			
HF	plays	7	VD	plays	10
	wins	5		wins	3
	%WR	71.4%		%WR	30.0%
	invest	787		invest	1006
	return	1053		return	621
	profit	266		profit	-385
	%ROI	33.8%		%ROI	-38.3%
VF	plays	0	HD	plays	4
	wins	0		wins	1
	%WR	0.0%		%WR	25.0%
	invest	0		invest	403
	return	0		return	202
	profit	0		profit	-201

%ROI	0.0%	%ROI	-49.9%
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c)V119 and V123 ensemble consensus only – 6th June to 21st Jun 2026

	#Games	208			
	#Plays	18			
	%Games	8.7%			
	#Win	8			
	%WinRate	44.4%			
	invest	1871			
	return	1662			
	profit	-209			
	%ROI	-11.2%			
HF	plays	5	VD	plays	9
	wins	4		wins	3
	%WR	80.0%		%WR	33.3%
	invest	565		invest	903
	return	839		return	621
	profit	274		profit	-282
	%ROI	48.5%		%ROI	-31.2%
VF	plays	0	HD	plays	4
	wins	0		wins	1
	%WR	0.0%		%WR	25.0%
	invest	0		invest	403
	return	0		return	202
	profit	0		profit	-201
	%ROI	0.0%		%ROI	-49.9%

d)V119 and V123 ensemble strategy as pe Section 7 above – 6th June to 21st Jun 2026

	#Games	208			
	#Plays	34			
	%Games	16.3%			
	#Win	14			
	%WinRate	41.2%			
	invest	3606			
	return	2903			
	profit	-703			
	%ROI	-19.5%			
HF	plays	14	VD	plays	16
	wins	7		wins	6
	%WR	50.0%		%WR	37.5%
	invest	1593		invest	1610
	return	1473		return	1228
	profit	-120		profit	-382
	%ROI	-7.5%		%ROI	-23.7%
VF	plays	0	HD	plays	4
	wins	0		wins	1
	%WR	0.0%		%WR	25.0%
	invest	0		invest	403
	return	0		return	202
	profit	0		profit	-201
	%ROI	0.0%		%ROI	-49.9%

9. Using V119 Plays as a Filter for V123

Another interesting phenomenon we found was that, when V119 and V123 both have a play but disagree, then the V123 appears extremely profitable, whilst the V119 plays are not – why is this?. The results are given below, including for Jun 2025.

a)V119 ensemble outsample plays that disagree with V123

#plays	37
%totalGames	42.5%
#wins	16
%winRate	43.2%
uInvestd	37.00
uRetnd	31.34
%ROI	-15.3%
Units	-5.66
SD(ROI)	98.4%
SR	
(outsample)	-0.94
SR	
(playbyplay)	-0.16

Fave and Dogs breakdown:

F			D		
	13.00	14.9%		24.00	27.6%
	11.44			19.90	
	-12.0%	-1.56		-17.1%	-4.10

b)V119 ensemble outsample plays that disagree with V123 – Season 2025

	2025
#plays	21
%SeasonGames	42.0%
%OS_Games	24.1%
#wins	9
%winRate	42.9%
uInvestd	21.00
uRetnd	17.76
%ROI	-15.4%
Units	-3.24

Fave and Dogs breakdown:

F	6.00	28.6%	D	15.00	71.4%
	5.75			12.00	
	-4.1%	-0.25		-20.0%	-3.00

c)V119 ensemble outsample plays that disagree with V123 – Jun 2025

Jun	
#plays	3
%totalGames	3.4%
#wins	1
%winRate	33.3%
uInvestd	3.00
uRetnd	1.97
%ROI	-34.3%
Units	-1.03

Fave and Dogs breakdown:

F	0.00	0.0%	D	3.00	100.0%
	0.00			1.97	
	0.0%	0.00		-34.3%	-1.03

d)V119 ensemble outsample plays that disagree with V123 – Season 2026

2026	
#plays	16
%SeasonGames	43.2%
%OS_Games	18.4%
#wins	7
%winRate	43.8%
uInvestd	16.00
uRetnd	13.59
%ROI	-15.1%
Units	-2.41

Fave and Dogs breakdown:

F	7.00	43.8%	D	9.00	56.3%
	5.69			7.90	
	-18.7%	-1.31		-12.2%	-1.10

e)V123 ensemble outsample plays that disagree with V119

#plays	82
%totalGames	94.3%
#wins	49
%winRate	59.8%
uInvestd	82.00
uRetnd	98.95
%ROI	20.7%
Units	16.95
SD(ROI)	99.9%
SR	
(outsample)	1.87
SR	
(playbyplay)	0.21

Fave and Dogs breakdown:

F	20.00	23.0%	D	62.00	71.3%
	25.00			73.94	
	25.0%	5.00		19.3%	11.94

f)V123 ensemble outsample plays that disagree with V119 – Season 2025

	2025
#plays	47
%SeasonGames	94.0%
%OS_Games	54.0%
#wins	26
%winRate	55.3%
uInvestd	47.00
uRetnd	52.90
%ROI	12.6%
Units	5.90

Fave and Dogs breakdown:

F	12.00	25.5%	D	35.00	74.5%
	15.48			37.43	
	29.0%	3.48		6.9%	2.43

g)V123 ensemble outsample plays that disagree with V119 – Jun 2025

Jun	
#plays	13
%totalGames	14.9%
#wins	5
%winRate	38.5%
uInvestd	13.00
uRetnd	10.19
%ROI	-21.6%
Units	-2.81

Fave and Dogs breakdown:

F	3.00	23.1%	D	10.00	76.9%
	3.87			6.32	
	29.0%	0.87		-36.8%	-3.68

h)V123 ensemble outsample plays that disagree with V119 – Season 2026

2026	
#plays	35
%SeasonGames	94.6%
%OS_Games	40.2%
#wins	23
%winRate	65.7%
uInvestd	35.00
uRetnd	46.04
%ROI	31.6%
Units	11.04

Fave and Dogs breakdown:

F	8.00	22.9%	D	27.00	77.1%
	9.53			36.52	
	19.1%	1.53		35.3%	9.52

MLB Scion — binary home-win classification. V119 (line + stats, 15 models) and V123 (stats-led, line removed, 41 models). SHAP aggregated across 14 V119 and 41 V123 models; stability tiers from rank SD across models. In-sample 2004–05 Jun 2025; out-of-sample 06 Jun 2025–05 Jun 2026 (1,770 games).